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pumpkingzy.github.io (主页有讲义, 课前若干天更新)

课后答疑是, 或者邮件预约时间

提交作业说明: 1. 课可以不来, 作业必须交.

2. 习题课可以任选助教听课.

3. 补交作业时限为下课后 7×24 小时.

课次 (6次): 3.9 3.23 4.20 5.9 5.18 6.1

课程内容概述:

线性空间理论 { 对象: 线性空间 (定义, 性质, 构造) } 和
{ 像, 核
张量积, 对称积, 外积
态射: 线性映射 (定义, 性质) (转化: 线性映射 $\xrightarrow{\text{基}}$ 矩阵)

Q: 能否找到一个基, 使得表示矩阵较简单 (对角? 上三角? 分块对角?)?

线性映射的标准型 { 工具: 多项式理论, 特征值
Jordan 标准型
有理标准型

额外结构 { 双线性型 $\leadsto$ 内积 (度量) $\leadsto$ 正交变换
Hermite 内积 (复内积) $\leadsto$ 酉变换
辛形式

作业题:

1B.6 $\mathbb{R} \cup \{\infty, -\infty\}$ 上定义了加法, $\mathbb{R}$ -数乘. 问它是不是一个 $\mathbb{R}$ -线性空间.

不是. 加法公理: 结合律 $(1+\infty)+(-\infty) = \infty + (-\infty) = 0$

$$1 + (\infty + (-\infty)) = 1 + 0 = 1$$

2A.11 $v_1, \dots, v_m$ 线性无关 $w_1, \dots, w_m$ 线性无关 $\Rightarrow v_1+w_1, \dots, v_m+w_m$ 线性无关.

反例. $V = \mathbb{R}^2$ $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $w_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ $w_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

2A.14 $v_1, \dots, v_m \in V$ $w_k = v_1 + \dots + v_k$ ($k \in \{1, 2, \dots, m\}$)

求证: $v_1, \dots, v_m$ 线性无关 iff $w_1, \dots, w_m$ 线性无关.

证明: " $\Rightarrow$ " $0 = \sum_{k=1}^m \lambda_k w_k = \sum_{k=1}^m \lambda_k \left(\sum_{j=1}^k v_j \right) = \sum_{j=1}^m \left(\sum_{k=j}^m \lambda_k \right) v_j$

则 $\sum_{k=j}^m \lambda_k = 0$ ($\forall j$) $j=m$ 时 $\lambda_m = 0$

$j < m$ 时 $\lambda_j = \sum_{k=j}^m \lambda_k - \sum_{k=j+1}^m \lambda_k = 0$

" $\Leftarrow$ " $0 = \sum_{k=1}^m \mu_k v_k = \mu_1 w_1 + \sum_{k=2}^m \mu_k (w_k - w_{k-1}) = \sum_{k=1}^{m-1} (\mu_k - \mu_{k+1}) w_k + \mu_m w_m$

则 $\mu_m = 0$ $\mu_k = \mu_{k+1}$ ($k < m$) 即 $\mu_k = 0$ ($\forall k$).

推广: 在上题中 $w_i = \sum_{j=1}^m a_{ij} v_j$, 若 $(a_{ij}) \in M_m(K)$ 是可逆方阵, 则结论仍成立.

证明: " $\Rightarrow$ " $0 = \sum_{i=1}^m \lambda_i w_i = \sum_{i,j} \lambda_i a_{ij} v_j$ 则 $\sum_{i=1}^m a_{ij} \lambda_i = 0$

$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} = 0$ 系数阵 $= (a_{ij})^t$ 可逆, 故 $\lambda_i = 0$.

" $\Leftarrow$ " (a_{ij}) 可逆. 则 $\exists (b_{ij}) \in M_m(K)$ s.t. $\sum_{i=1}^m b_{ki} a_{ij} = \delta_{kj}$ 且 (b_{ij}) 可逆

$$\sum_{i=1}^m b_{ki} w_i = \sum_{i,j} b_{ki} a_{ij} v_j = \sum_{j=1}^m \delta_{kj} v_j = v_k$$

所以由 " $\Rightarrow$ " 结论即可证明 $v_1, \dots, v_m$ 线性无关.

例1. (23春 线代AII 期中T1) (改)

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \in M_2(\mathbb{R}) \quad A: M_2(\mathbb{R}) \longrightarrow M_2(\mathbb{R})$$
$$X \longmapsto AX$$

(1) 求恒 A 为线性变换. (2) 求 $\text{Im } A$ 与 $\ker A$ 的基与维数.

(3) 将 $\text{Im } A$ 的基扩充成 $M_2(\mathbb{R})$ 的基.

解: (1) 略. (2) $\text{Im } A$ 有生成元 $\{AE_{ij}\}_{1 \leq i, j \leq 2} = \left\{ \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 4 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 0 & 4 \end{pmatrix} \right\}$

极大无关组 $\left\{ \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix} \right\}$, 故 $\dim \text{Im } A = 2$.

$$X \in \ker A \Leftrightarrow AX = 0 \quad X = (v_1, v_2) \quad Av_1 = 0, Av_2 = 0 \Rightarrow v_1, v_2 \in \text{span} \left\{ \begin{pmatrix} 2 \\ -1 \end{pmatrix} \right\}$$

$\ker A$ 有基 $\left\{ \begin{pmatrix} 2 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \right\}$, 故 $\dim \ker A = 2$.

(3) $M_2(\mathbb{R})$ 有标准基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 后扩充入得 $\left\{ \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, E_{11}, E_{22} \right\}$.

例2. (24春 线代AII 期中T1)

$$K^4 \{e_i\}. \quad \text{令 } V_1 = \{(x, y, z, t) \in K^4 \mid 2x + y + z - t = 0, x + y + z = 0\}$$

$$V_2 = \text{span} \{e_1 + e_3\} \quad V_3 = \text{span} \{e_1 + e_2\}$$

判断 $V_1 + V_2 + V_3$ 是否为直和, 并计算 $\dim(V_1 + V_2 + V_3)$.

解: $\begin{cases} t = 2x + y + z = x \\ z = -x - y \end{cases} \quad V_1 \text{ 有基 } \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad V_1 + V_2 + V_3 \text{ 生成元拼成矩阵}$

$$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad r=3$$

$$\text{从而 } \dim(V_1 + V_2 + V_3) = 3.$$

例3. (24春 线代AII 期中 T6)

$V_1, V_2 \subseteq V$ 子空间 $V = V_1 + V_2$ 令 W_1 为 $V_1 \cap V_2$ 在 V_1 中的补空间.

求证: $V = W_1 \oplus V_2$.

证明: $\forall v \in V \quad v = v_1 + v_2 \quad (v_1 \in V_1, v_2 \in V_2)$ 由 $V_1 = (V_1 \cap V_2) \oplus W_1$

$$v_1 = w_1 + v_0 \quad (w_1 \in W_1, v_0 \in V_1 \cap V_2) \quad v = w_1 + (v_0 + v_2) \in W_1 + V_2$$

$$\forall d \in W_1 \cap V_2 \Rightarrow d \in W_1 \subseteq V_1 \Rightarrow d \in V_1 \cap V_2 \text{ 又 } d \in W_1 \text{ 故 } d = 0.$$

例4. (24春 线代B 期末 T7) 设 $A, B, C, D \in M_n(K)$ 两两可交换. $AC + BD = I$.

$$\text{设 } V = \{x \in K^n : ABx = 0\} \quad V_1 = \{x \in K^n : Bx = 0\} \quad V_2 = \{x \in K^n : Ax = 0\}$$

求证: $V = V_1 \oplus V_2$.

证明: $V = V_1 + V_2$: $\forall v \in V \quad v = ACv + BDv$ 其中 $ACv \in V_1, BDv \in V_2$.

$$V_1 \cap V_2 = \{0\}: \forall v \in V_1 \cap V_2 \text{ 则 } v = ACv + BDv = CAv + DBv = 0 + 0 = 0.$$

例5. (25春 线代AII 期末 T6)

设 $V = \mathbb{R}[x]_{\leq n}$. (1) 求证 $1, (x+1), (x+1)^2, \dots, (x+1)^n$ 是 V 的基.

(2) 求 $D: V \rightarrow V$, 求 D 在 (1) 中基下的表示阵.

(3) 求 D 的所有不变子空间. (即 $W \subseteq V$ 子空间满足 $DW \subseteq W$)

解: (1) 略 (2) $D(x+1)^k = k(x+1)^{k-1} \quad (k \geq 1) \quad D(1) = 0$

$$\text{所以 } (D1, D(x+1), \dots, D(x+1)^n) = (1, x+1, \dots, (x+1)^n) \begin{pmatrix} 0 & 1 & & & \\ & 0 & 2 & & \\ & & 0 & \ddots & \\ & & & 0 & n \\ & & & & 0 \end{pmatrix}$$

(3) 已知 $\mathbb{R}[x]_{\leq k}$ 都是不变子空间 ($k=0, 1, \dots, n$). 下证只有这些不变子空间.

取 W 是不变子空间. 记 $k = \max_{f \in W} \deg f$. 下证 $W = \mathbb{R}[x]_{\leq k}$

首先 " $\subseteq$ " 是显然的. 再证 " $\supseteq$ ". $\Rightarrow f(x) = x^k + a_{k-1}x^{k-1} + \dots + a_1x + a_0 \in W$

$$\text{则 } D^k f = k! \in W \Rightarrow 1 \in W. \quad D^{k-1} f = \frac{k!}{1!} x + a_{k-1}(k-1)! \in W \Rightarrow x \in W$$

类似地可得 $1, x, \dots, x^k \in W$, 则 $\mathbb{R}[x]_{\leq k} \subseteq W$. ▣

例6. (Wronsky 行列式) 令 $V = C^\infty(\mathbb{R})$. $f_1, f_2, \dots, f_n \in V$. 记 $W = \det(f_j^{(i-1)}) \in V$.

若 $W \neq 0 \in V$, 则 $f_1, \dots, f_n$ 在 V 中线性无关.

证明: $\sum_{k=1}^n \lambda_k f_k = 0$ 两边对 x 求导 n 次得 $\sum_{k=1}^n f_k^{(i-1)} \lambda_k = 0 \quad (i=1, 2, \dots, n)$

系数阵 $(f_j^{(i-1)})$ 取 x 使 $W(x) \neq 0$, 则系数阵可逆, 故 $\lambda_k = 0 \quad (k=1, 2, \dots, n)$ ▣

应用: $e^{\lambda_1 x}, \dots, e^{\lambda_n x} \in C^\infty(\mathbb{R})$, 若 λ_i 两两不同, 则它们线性无关.

证明: $W(x) = e^{(\lambda_1 + \dots + \lambda_n)x} \begin{vmatrix} 1 & \dots & 1 \\ \lambda_1 & \dots & \lambda_n \\ \vdots & & \vdots \\ \lambda_1^{n-1} & \dots & \lambda_n^{n-1} \end{vmatrix} = e^{(\sum \lambda_i)x} \prod_{i < j} (\lambda_j - \lambda_i) \neq 0$. ▣

例7. (24秋线代B 期末T1)

V 是一个 n 维线性空间, $A, B: V \rightarrow V$ 线性映射, 判断:

(1) $r(A) = r(B) \Rightarrow r(A^2) = r(B^2)$ 反例: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

(2) $r(A) + r(B) \geq r(A+B)$ 证明: $\text{Im}(A+B) \subseteq \text{Im} A + \text{Im} B$

从而 $\dim \text{Im}(A+B) \leq \dim \text{Im} A + \dim \text{Im} B - \dim(\text{Im} A \cap \text{Im} B) \leq \dim \text{Im} A + \dim \text{Im} B$.

例8. (24秋线代B 期末T6) 令 $V = \{A \in M_2(\mathbb{R}) \mid \text{tr} A = 0\}$, 取 $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ 定义映射

$$\begin{aligned} \Phi_P: V &\longrightarrow V \\ A &\longmapsto P^{-1}AP \end{aligned} \quad \begin{aligned} (1) &\text{ 求证 } V \text{ 有基 } \{E_{11}-E_{22}, E_{12}, E_{21}\}. \\ (2) &\text{ 求 } \Phi_P \text{ 在这组基下的表示矩阵}. \end{aligned}$$

解: $\Phi_P \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Phi_P(E_{12}) = E_{21} \quad \Phi_P(E_{21}) = E_{12}$

所以 Φ_P 的表示矩阵是 $T = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. 分块对角

(3) 求 Φ_p 的特征值与特征向量.

解: $p_{\Phi_p}(\lambda) = \det(\lambda I_3 - T) = \begin{vmatrix} \lambda+1 & & \\ & \lambda & -1 \\ & -1 & \lambda \end{vmatrix} = (\lambda+1)(\lambda^2-1)$

$\lambda=1$ 的特征向量 $E_{12}+E_{21}$

$\lambda=(-1)$ 的特征向量 $E_{11}-E_{22}, E_{12}-E_{21}$.

(4) 对 $U \in M_2(\mathbb{R})$ 类似可以定义 $\Phi_U(A) = U^T A U$. 若 U 有特征值 2 和 4,

求 Φ_U 的特征值.

解: 存在可逆 $Q \in M_2(\mathbb{R})$ 使 $Q^T U Q = D = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$. $U = Q D Q^{-1}$

则 $\Phi_U(A) = U^T A U = Q D^T Q^{-1} A Q D Q^{-1} = \Phi_Q^{-1} \circ \Phi_D \circ \Phi_Q(A)$

作为 V 上的线性变换, Φ_U 与 Φ_D 相似, 有相同的特征值.

$$\Phi_D \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \\ & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \begin{pmatrix} 2 & \\ & 4 \end{pmatrix} = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \quad \lambda=1$$

$$\Phi_D \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \quad \lambda=2$$

$$\Phi_D \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \frac{1}{2} & 0 \end{pmatrix} \quad \lambda=\frac{1}{2}$$

例9. (25春线代II 期中T7) $A \in M_n(\mathbb{C})$ 有 n 个不同的特征值.

考虑线性变换 $L: M_n(\mathbb{C}) \longrightarrow M_n(\mathbb{C}) \quad B \longmapsto A B A^T$

(1) 设 X, Y 是 A 属于 λ_i, λ_j 的特征向量. 构造一个 L 的特征向量.

(2) 求一个 L 的非平凡零化多项式.

解: (1) $A X = \lambda_i X, A Y = \lambda_j Y \Rightarrow Y^t A^t = \lambda_j Y^t$

则 $L(X Y^t) = A X Y^t A^t = \lambda_i X \cdot \lambda_j Y^t = (\lambda_i \lambda_j) \cdot X Y^t$. (为什么 $X Y^t \neq 0$?)

(2) 设 $A = P^{-1} D P$ $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, 令线性变换 $\varphi(B) = P B P^t$

则 $\varphi^{-1}(B) = P^t B P^{-1}$, 则有 $\varphi \circ L \circ \varphi^{-1}(B) = P(A(P^t B P^{-1})A^t)P^t = P D B D^t$. 记为 $\tilde{L}$.

Define $\tilde{L}(b_{ij}) = (\lambda_i \lambda_j b_{ij})_{n \times n}$ for fixed $(\lambda_i \lambda_j, E_{ij})$, has characteristic polynomial $p(x) = \prod_{i,j} (x - \lambda_i \lambda_j)$.
 $\lambda \in L$ and $\tilde{L}$ has rank ≤ 1 $\Rightarrow p(x)$ is L -monic polynomial. $\square$

Exercise:

3F.6 $\varphi, \beta \in V'$ Show that $\text{null } \varphi \subseteq \text{null } \beta \iff \exists c \in F \text{ s.t. } \beta = c\varphi$.

p.f. " $\Leftarrow$ " $\forall v \in \text{null } \varphi \quad \beta v = c \cdot \varphi v = c \cdot 0 = 0 \Rightarrow v \in \text{null } \beta$

" $\Rightarrow$ " $\text{Im } \beta \subseteq F \Rightarrow \text{rank } \beta \leq 1$ If $\text{rank } \beta = 0$, then $\beta = 0$, pick $c = 0$.

If $\text{rank } \beta = 1 \Rightarrow \dim \text{null } \beta = \dim V - 1 \geq \dim \text{null } \varphi = \dim V - \text{rank } \varphi \geq \dim V - 1$

$\Rightarrow \text{null } \varphi = \text{null } \beta$ fix a basis $\{v_1, v_2, \dots, v_{n-1}\}$ $n = \dim V$, extend to $\{v_1, \dots, v_{n-1}, w\}$

$\Rightarrow \varphi, \beta \in \text{span } \{w^*\} \setminus \{0\} \Rightarrow \exists c \in F \text{ s.t. } \beta = c\varphi$.

3F.20 V finite dimensional, $U \subseteq V$. Show that $U = \{v \in V : \varphi(v) = 0, \forall \varphi \in U^\circ\}$.

p.f. U admits a basis $\{u_1, \dots, u_k\}$, extend to $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$, a basis of V .

U° admits a basis $\{v_{k+1}^*, \dots, v_n^*\}$. $U \subseteq \{v \in V : \varphi(v) = 0, \forall \varphi \in U^\circ\} \checkmark$

" $\supseteq$ " $\forall w \in \text{RHS} \quad w = \lambda_1 u_1 + \dots + \lambda_k u_k + \mu_{k+1} v_{k+1} + \dots + \mu_n v_n$

$v_j^* w = 0 \Rightarrow \mu_j = 0 \quad (k+1 \leq j \leq n) \Rightarrow w \in U$.

3F.28 V finite dimensional, $\varphi_1, \dots, \varphi_m$ is linearly independent in V' .

Show that $\dim((\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m)) = \dim V - m$.

p.f. Extend $\{\varphi_1, \dots, \varphi_m\}$ to $\{\varphi_1, \dots, \varphi_m, \varphi_{m+1}, \dots, \varphi_n\}$, a basis of V' . $V'' \cong V$.

$\text{null } \varphi_i$ admits a basis $\{\varphi_1^*, \dots, \widehat{\varphi_i^*}, \dots, \varphi_m^*, \varphi_{m+1}^*, \dots, \varphi_n^*\}$

It just remains to show that $(\text{null } \varphi_1) \cap \dots \cap (\text{null } \varphi_m) = \text{span}\{\varphi_{m+1}^*, \dots, \varphi_n^*\}$.